Three Ways to Measure a Body

On the same scan the pipelines return chest circumferences **80 cm** apart. None is malfunctioning. Each answers a different question about the same mesh, and the answers cost between 0.6 and 7 seconds. This is what each one assumes, what that buys, and what it costs.

What each pipeline calls a circumference

AVATAR SEGMENTATION SLICE

Everything below follows from one choice: at height z, which curve are you measuring the length of?

AVATAR — HULL OF A BAND

$$G_{\mathrm{av}}(z) = |\partial \mathrm{conv}\, \pi(B_z)|$$

Not a cross-section. B_z is every vertex of every face that straddles the plane — points above and below alike. They are projected to the plane (π) and the convex-hull perimeter is reported. Nothing is interpolated, so the girth is read off a slab, not a slice.

SLICE — SUM OF EVERY LOOP

$$G_{\mathrm{sl}}\left(z\right) = \sum_{k=1}^{K(z)} |C_k(z)|$$

A true planar cross-section, giving $K(z)$ closed loops — at chest height usually three: one torso, two arms. The reported girth is their sum. The per-loop lengths are computed and written to disk; only the aggregation collapses them.

SEGMENTATION — REGION FIRST

$$G_{\mathrm{sg}}\left(z\right) = |\partial(M_{\mathrm{trunk}} \cap P_z)|$$

The mesh is cut into anatomical regions first, so the section P_z is taken of the trunk alone and the arms never enter it. It solves directly what the other two work around, and pays for it in time.

Avatar: hull of a band, and what that costs

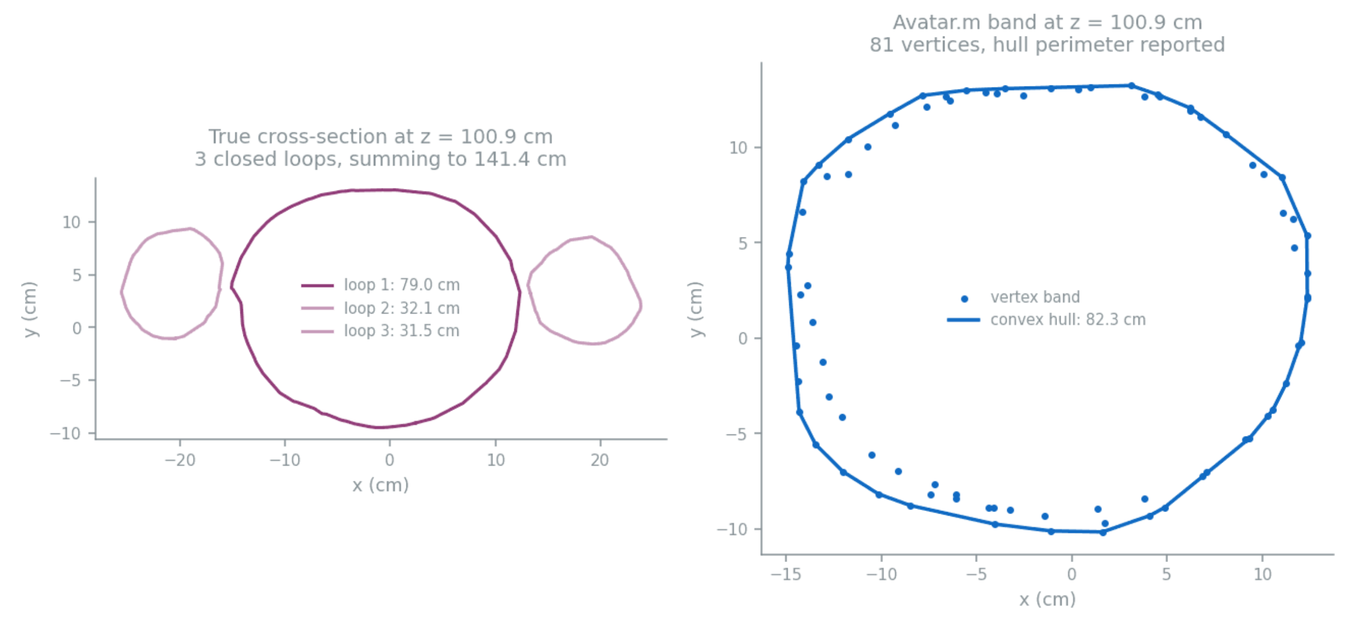
AVATAR

The band-and-hull definition has two effects that pull in opposite directions, and they separate exactly. Writing C for the true torso loop on the same plane:

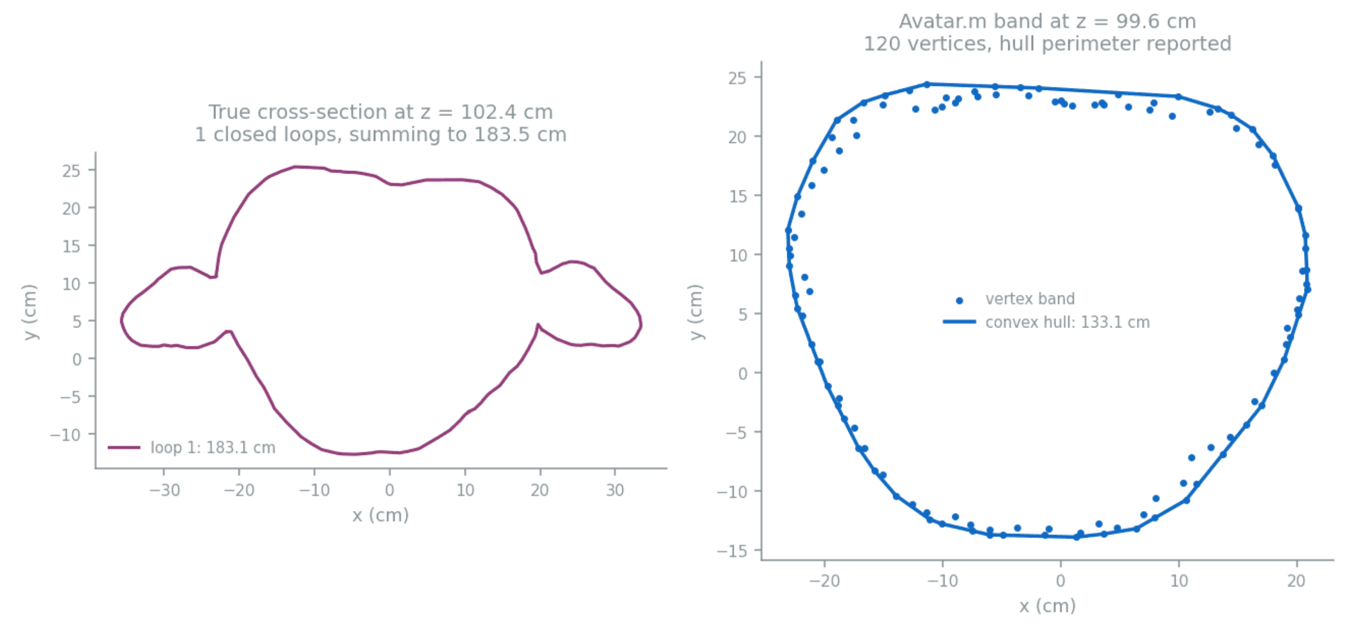
$$\underbrace{G_{\mathrm{av}} - |C|}_{\text{NET OFFSET}} = \underbrace{\left(G_{\mathrm{av}} - |\partial \mathrm{conv}\, C|\right)}_{\text{BAND WIDENING}} + \underbrace{\left(|\partial \mathrm{conv}\, C| - |C|\right)}_{\text{HULL SHORTCUT}}$$

The second term is **never positive**. The convex hull is the shortest closed curve enclosing a point set, so $|\partial \mathrm{conv}\, C| \le |C|$ for every closed C. Convexity can only pull a girth *down*. Anything pushing it up came from the band — a slab of finite thickness through a tapering body, whose projection is wider than any single section inside it.

Over 45 sections that separate into distinct loops: band widening **+2.57 cm**, hull shortcut **-0.55 cm**, net **+2.02 cm**. The band dominates. Worth stating precisely, because the intuition runs the other way — a convex hull is usually assumed to over-read a waist, and here that is not where the excess comes from.



Chest height on A00-08-4914_B. **Left:** the true cross-section is three separate closed loops — torso 79.0 cm, arms 32.1 and 31.5 cm. The slice pipeline adds all three and reports 141.4 cm. **Right:** Avatar.m collects a band of nearby vertices and takes the convex hull, reporting 82.3 cm — the torso, plus about 3 cm of convexity where the hull cuts across the armpit hollows.



The same height on CanCan01_A, where the arms rest against the torso. Now there is only *one* loop: the cross-section walks out around each arm and back, so the sum and the largest loop are the same 183.5 cm. Avatar.m's hull bridges straight across both armpits and returns 133.1 cm. No aggregation rule recovers the torso here — the loops would have to be cut apart first.

The second figure is the case no aggregation rule survives. With the arms against the torso the section is a *single* loop that already contains them, so $K(z) = 1$ and the sum and the largest loop are both 183.5 cm. The hull is indifferent — it bridges the armpits either way. That indifference is what buys Avatar.m its robustness, and what makes it approximate.

Constants or search: how each finds a level AVATAR SEGMENTATION SLICE

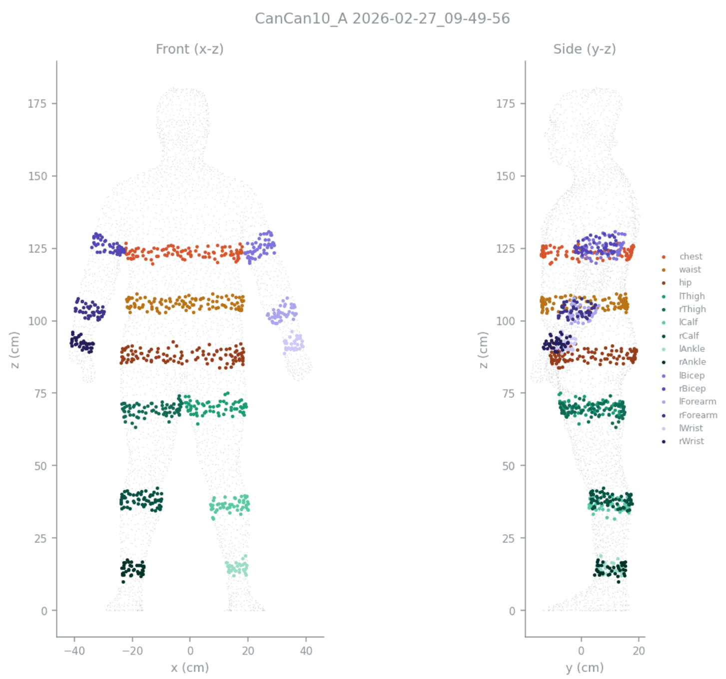
The second axis: how each pipeline decides *where* a level sits. A fixed fraction of stature is fast and reproducible and wrong on atypical bodies; a geometric search adapts, and can fail outright.

LEVEL	AVATAR	SEGMENTATION	SLICE
chest	CONSTANT	HYBRID	HYBRID
waist	CONSTANT	HYBRID	HYBRID
hip	HYBRID	HYBRID	HYBRID
thigh	CONSTANT	CONSTANT	HYBRID
crotch	SEARCH	SEARCH	HYBRID
arm	HYBRID	HYBRID	CONSTANT
orientation	90°/180° axis permutation by bounding-box extent. Never a free rotation, so stature equals a bbox extent exactly.	Rule-based whole-body axis pick, then *per-limb* OBB alignment after segmentation.	True PCA on the vertex covariance. No azimuthal correction, so left/right is inherited from the file.
limb separation	Arms: constrained flood fill from the fingertip. Legs: *no connectivity at all* — two cutting lines, crotch to each hip.	Boolean subtraction, plane cut, connected components scored by shape, with a geodesic flood-fill fallback.	**None.** Every value is read off whole-body slice statistics inside a stature band.
tally	5 constant · 3 search · 7 hybrid	4 constant · 3 search · 14 hybrid	4 constant · **0 search** · 13 hybrid

Hover any cell for the criterion and the source line. **constant** = a fixed fraction or midpoint, no geometry consulted. **search** = an unrestricted extremum, derivative or convexity criterion. **hybrid** = a search boxed inside a hard-coded window.

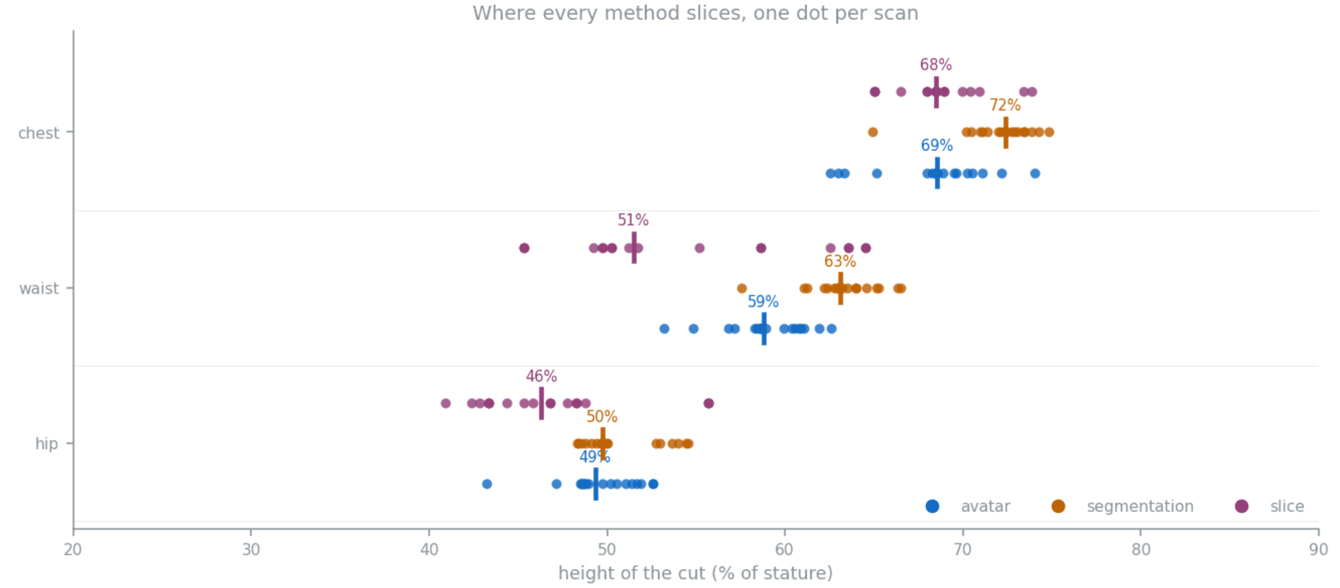
The ordering is not what the accuracy table implies. *Segmentation* is the most search-driven — its crotch and armpit detection run real optimisations with no fixed finish line. *Slice* is the most constant-driven: no unrestricted search anywhere, every extremum boxed inside a percent-of-stature window that supplies most of the anatomical information, and two measurements that are pure fractions of height. *Avatar* sits between, but is constant-heavy exactly where it matters most: chest and waist are a median and a midpoint of armpit and hip heights, with no fullness or narrowing search at all.

That is the real trade. A constant is reproducible, cheap, and cannot fail loudly; it is simply wrong on any body it was not fitted to. A search adapts to the body in front of it and can fail outright — which is what the outlier scan below shows happening.

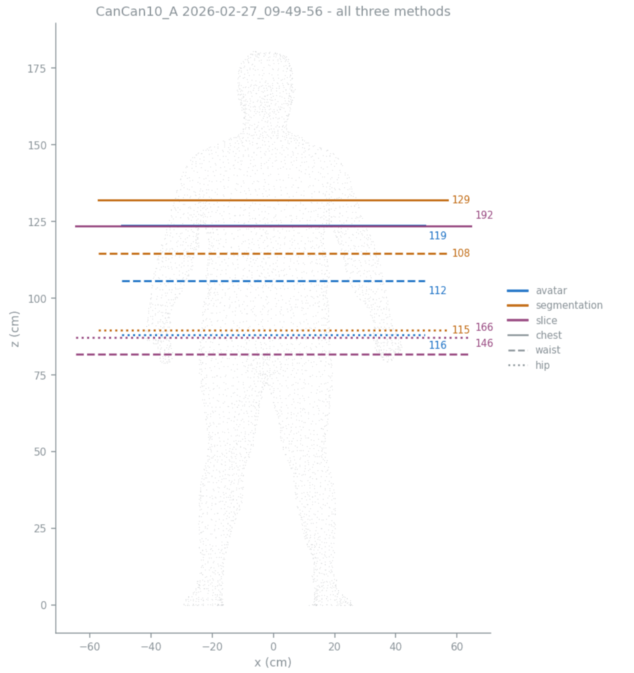


Every girth **Avatar.m** measures on CanCan10_A, drawn on the scan it came from. Each colour is one measurement; the points are the vertices that went into it. Limb girths sit at an angle because they are cut perpendicular to the limb axis, not to the floor.

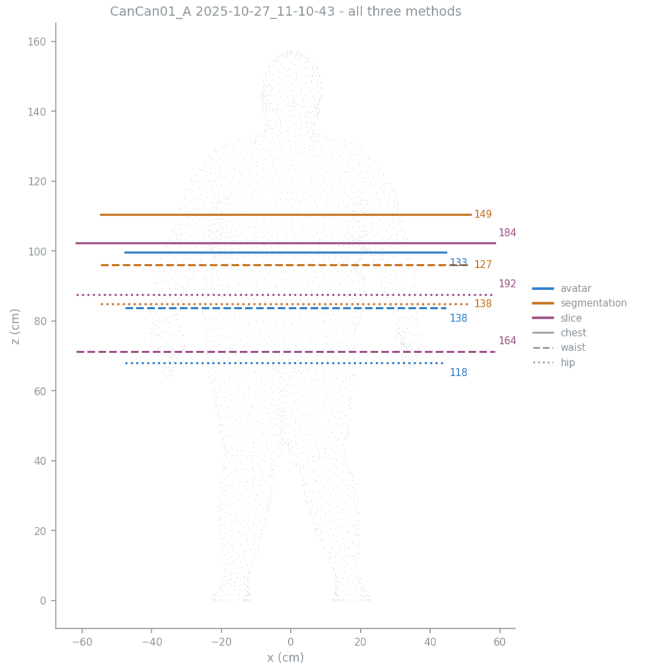
Where each pipeline cuts AVATAR SEGMENTATION SLICE



The height of every cut, as a percent of stature, for all three methods across the cohort. Each dot is one scan; the bar is the median. Chest is the agreement case — avatar and slice cut within half a point of each other (69% and 68%), with segmentation 4 points higher because it targets the bust rather than the median of the armpits. Waist is the disagreement case: three methods, three levels (51%, 59%, 63%), and slice's spread is 5× the others (IQR 11.8 points against 2.4). The three demo meshes shipped for smoke tests are excluded: they sit at a different scale and are not body scans.

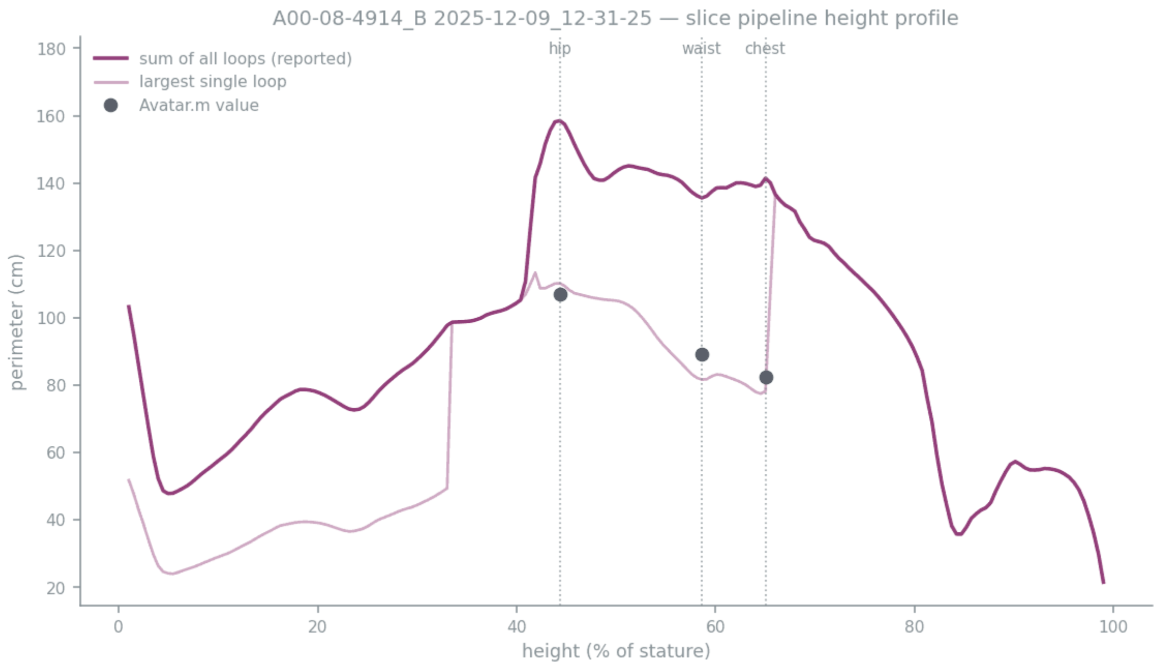


All three methods' chest, waist and hip on CanCan10_A, with the value each reports. Avatar and slice agree on chest *height* to within a point, so the 119 against 192 is definition alone. Segmentation cuts higher, at the bust. Slice's waist sits below its own hip.



CanCan01_A, the largest scan in the folder. Here the methods disagree about height as well as definition, and the slice waist again lands below the slice hip.

On CanCan10_A the chest heights coincide, which isolates that disagreement to definition alone. It does not hold everywhere — on CanCan01_A the heights differ too. Waist and hip fail differently again: on 5 of 20 scans the slice pipeline places its waist at or below its hip, an ordering problem upstream of any measurement.

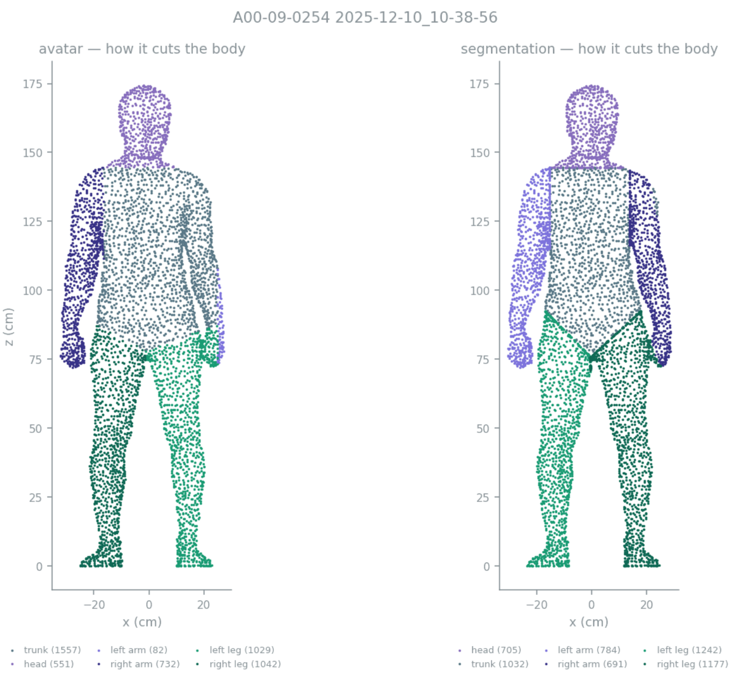


The slice pipeline's own height profile for A00-08-4914_B, which it computes and writes out in full. The reported curve is the sum of every loop; the faint curve is the largest single loop. Avatar.m's values sit on the faint curve, not the reported one — the number the pipeline needs is already in the file it wrote.

Avatar vs segmentation: when the reference is the one that fails

MATLAB AVATAR SEGMENTATION

One scan looks like an outlier for every method — segmentation 128%, slice 180%, and the only scan where the port and MATLAB disagree. It is worth being careful about what that means, because the scan is fine and so is segmentation.

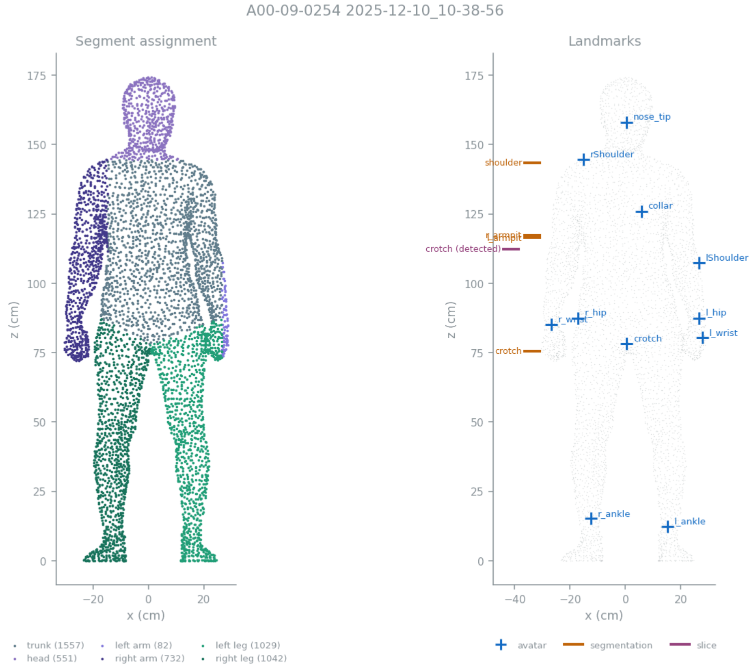


The same mesh, cut up by each backend, drawn at the same true scale. **Left:** `Avatar.m` — one arm segment holds 82 vertices, a sliver at the hand, and the trunk absorbs the rest of that arm at 1557 vertices. **Right:** the segmentation backend separates both arms, 784 and 691, leaving a trunk of 1032 — a trunk rather than a trunk-plus-arm. Note also that the two label left and right oppositely: avatar puts 'left' at x > 0, segmentation at x < 0. Swapping the sides back changes segmentation's paired left/right error only from 25.6% to 23.9%, so the convention is not what drives its numbers.

`Avatar.m`'s left-arm search collapses on this mesh. Its arm segment ends up with 82 vertices against the right arm's 732, and every landmark on that side follows: `lShoulder` lands 37 cm below `rShoulder`, and the reported left bicep is **4.9 cm** — a circumference of about 1.6 cm across, which is not an arm.

The mesh is not the problem. Segmentation's boolean-and-connectivity split separates both arms cleanly on the same file, 784 and 681 vertices, and returns arm lengths within 1% of each other:

ON THIS SCAN	MATLAB	AVATAR	SEGMENTATION
arm length, left / right (cm)	27.3 / 44.5	27.3 / 44.5	45.5 / 45.3
bicep, left / right (cm)	4.9 / 43.5	4.9 / 43.5	36.1 / 33.1
arm surface area, left / right (cm²)	295 / 2349	295 / 2349	2146 / 1790
left-right asymmetry	39%	39%	1%
arm segment, vertices	–	82 / 732	784 / 681



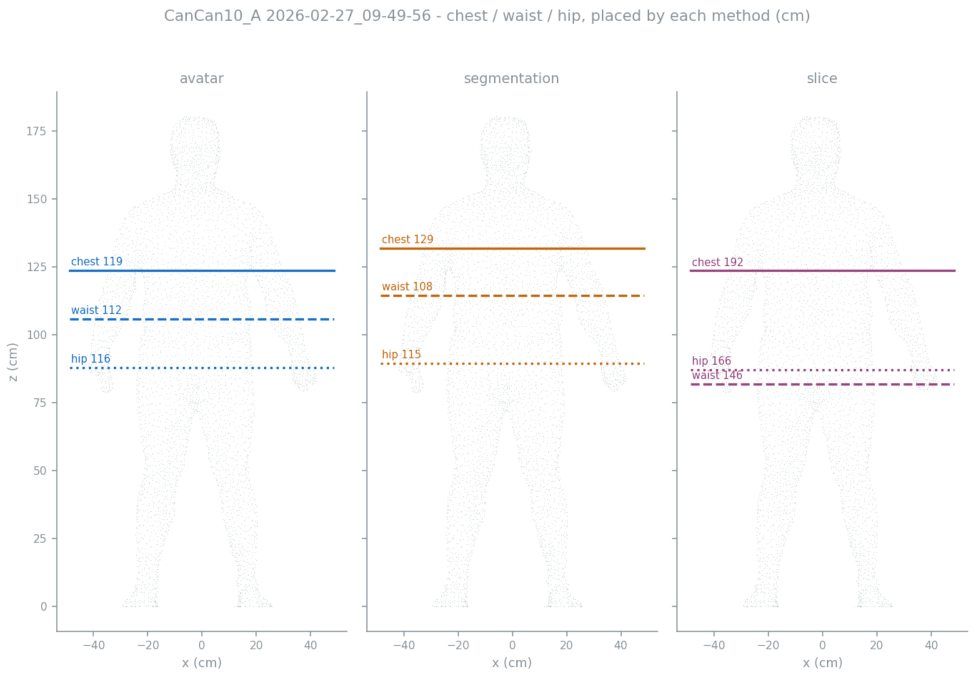
The landmarks that follow from those cuts, each method in its own colour. Segmentation and slice report in their own frames, so only height is comparable and they are drawn in lanes. Avatar's `lShoulder` sits 37 cm below its `rShoulder`; segmentation puts one shoulder height for both. Slice's crotch lands at chest height — its detector takes the highest level with two or more loops, which the separated arms satisfy all the way up to its band ceiling.

> **Segmentation's 128% on this scan is measured against a reference that is wrong here.** The port reproduces `Avatar.m` faithfully, including the failure, so both agree on a 4.9 cm bicep. Segmentation disagrees with them because it got the arm right.
>
> This also explains the one crotch the port cannot reproduce. `adjustCrotch` sweeps from the crotch up to the lower armpit, and it excludes arm vertices from that sweep — both inputs are corrupted by the same failed arm search. The notch profile it clusters comes out degenerate, which is why MATLAB's randomly seeded `kmeans` is unstable on this scan and no other.

The general point is that limb separation is the step every later measurement inherits, and the three backends are not equally exposed to it. Avatar cuts arms with a constrained flood fill and legs with two straight lines. Segmentation does boolean subtraction, a plane cut, connected-component scoring and a geodesic fallback — which is most of why it costs what it costs. Slice does not separate limbs at all, which is why its crotch detector, drawn above, finds the arms instead: it takes the highest level with two or more loops, and the torso plus two arms satisfies that all the way up to its band ceiling at 65% of stature.

Which placement is anatomically right AVATAR SEGMENTATION SLICE

Agreement with `Avatar.m` and agreement with anatomy are not the same ranking, and on chest, waist and hip they are close to opposite.



The same body, the same three girths, placed by each pipeline. The value each one reports is printed on its line. Read the ladder: segmentation puts the chest at the bust, the waist above the navel and the hip at the widest point. Slice puts its waist *below* its hip.

Two checks, neither of which uses `Avatar.m` as the reference.

Ordering. Chest must sit above waist, and waist above hip, on every body. Avatar gets this right on 18 of 18 scans and segmentation on 18 of 18. Slice manages 14 of 18, inverting waist and hip on 4 of them.

Absolute placement. Standard anthropometry puts the chest near 70–72% of stature, the natural waist near 62–64%, and the hip near 48–52%. Median placements:

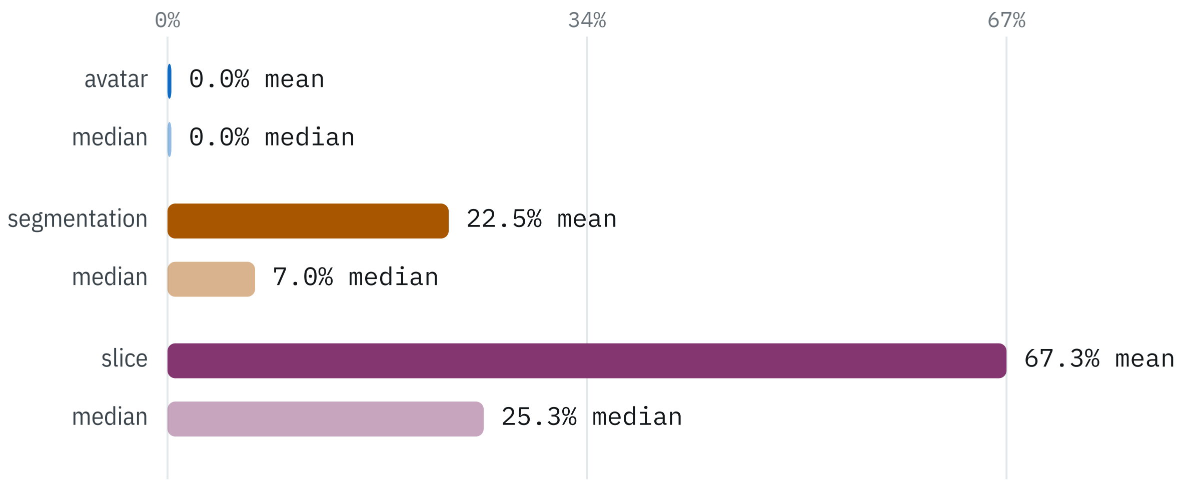
METHOD	CHEST	WAIST	HIP	WAIST IQR	MEAN MISS	ORDERED
expected	~71%	~63%	~50%	–	–	–
segmentation	**72%**	**63%**	**50%**	2.0	0.6	18/18
avatar	69%	59%	**49%**	2.4	2.4	18/18
slice	68%	51%	46%	11.8	5.9	14/18

Segmentation sits closest to the expected level on all three — a mean miss of 0.6 points of stature, against avatar 2.4, slice 5.9. Avatar's chest is 2 points low and its waist 4 points low, because both are midpoints of armpit and hip rather than searches for the bust or the narrowing. Slice's waist is 12 points low with 5× the spread.

This is the uncomfortable result. Segmentation scores worst against `Avatar.m` on exactly these measurements — chest 6.6%, waist 9.9% — and it is the one putting the tape in the right place. The gap is the reference's, not segmentation's.

How far each sits from Avatar.m AVATAR SEGMENTATION SLICE

Scored against `Avatar.m` as run in MATLAB R2023b — the reference because it is the implementation the others were written from, not because it is known to be right.



Distance from Avatar.m, mean and median. Only pairs where both produced a value are scored. This measures *distance from the reference*, not correctness — the placement test further down finds the ranking reverses on chest and waist. Hover a bar for the exact value.

The two statistics say different things about segmentation. Its mean is 22.5%, its median 7.0% — the mean is carried by a short list of columns rather than by typical behaviour. Left and right arm volume alone are **40% of its entire error budget**, and `collar_to_scalp_length` another 10%, which is the measurement where the reference is the one with the defect. Drop the segment volumes — a different quantity, computed from a hole-filling routine the port does not implement — and segmentation sits at 12.8% mean, 5.8% median.

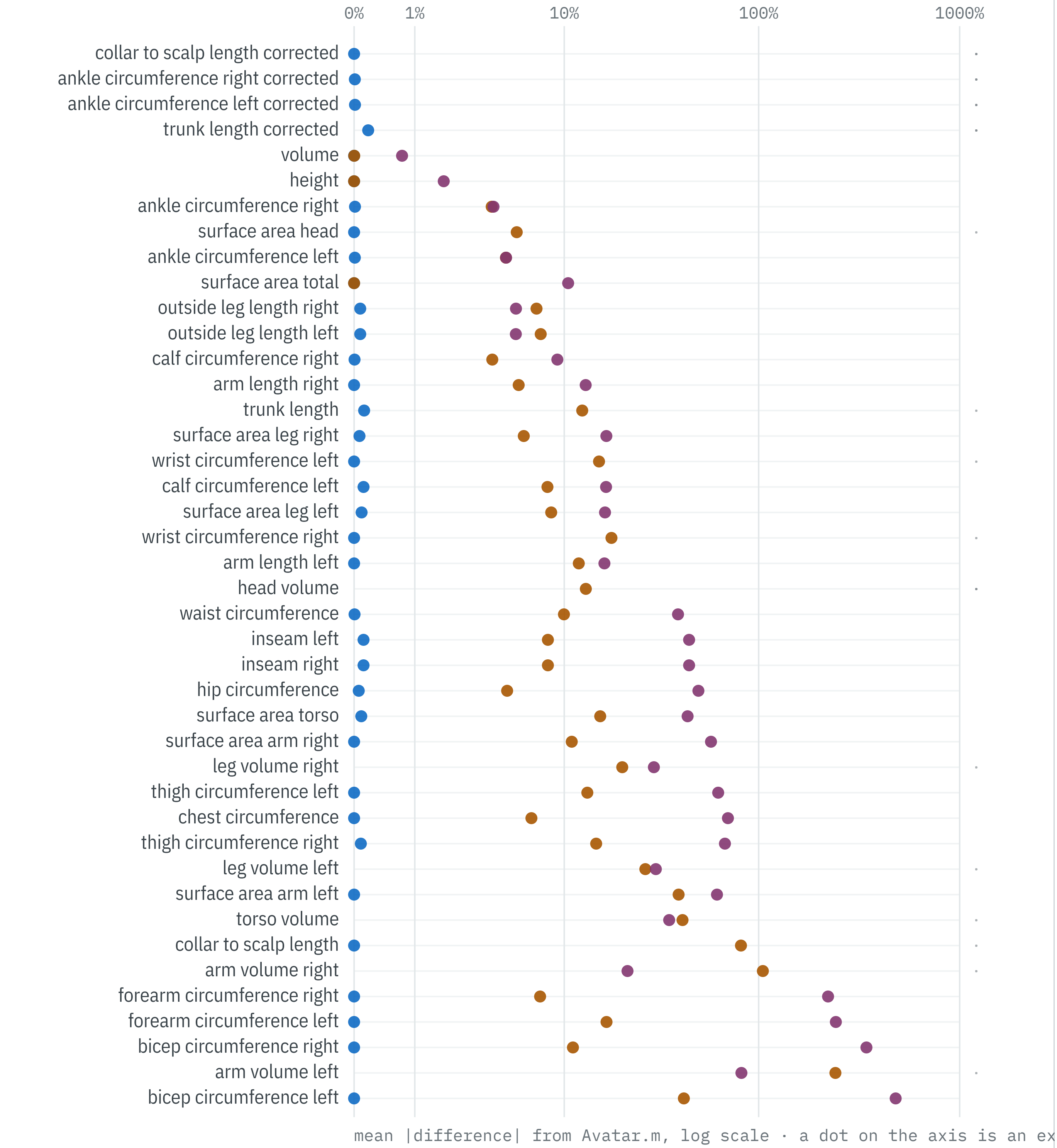
Slice does not have that structure. Its median is 25%, so the disagreement is the whole distribution rather than a tail.

EVERY PAIR, WITH NEITHER TREATED AS TRUTH

	MATLAB	AVATAR	SEGMENTATION	SLICE
matlab	–	0.03%	17.70%	37.93%
avatar	0.03%	–	12.91%	39.28%
segmentation	17.70%	12.91%	–	46.04%
slice	37.93%	39.28%	46.04%	–

Each cell is the mean difference between two methods as a percentage of their average, so it assumes neither is correct. Segmentation sits closer to the MATLAB pair (17.7%) than slice sits to anything (38% from MATLAB, 46% from segmentation).

EVERY MEASUREMENT, RANKED



Mean absolute difference from Avatar.m for every measurement the method and the reference both produce, sorted by average disagreement. Hover a dot for the exact value. Anything past 100% is clipped and marked; the segment volumes run into the thousands of percent, which is a different quantity being computed rather than a different answer.

THE MEASUREMENTS PEOPLE ASK FOR

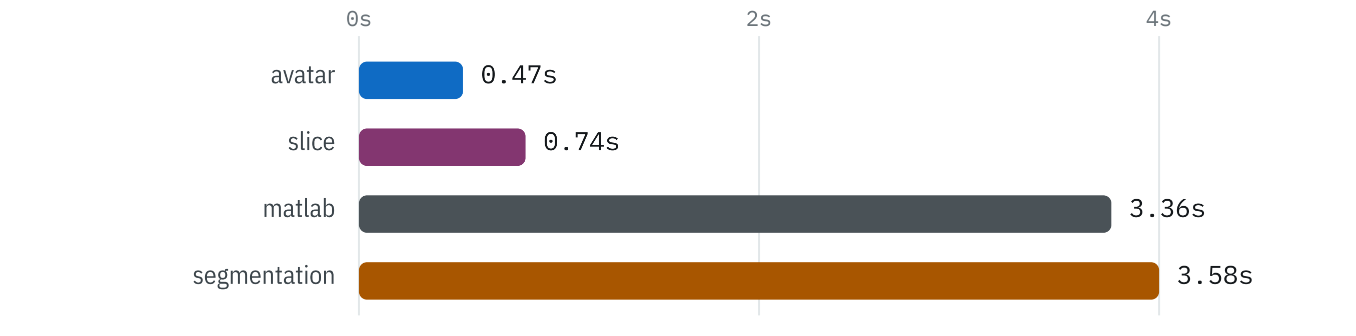
MEASUREMENT	AVATAR	SEGMENTATION	SLICE
height	0.00% · 20/20	0.00% · 20/20	1.78% · 0/20
chest circumference	0.00% · 20/20	6.56% · 0/20	70.11% · 0/20
waist circumference	0.01% · 19/20	9.95% · 0/20	39.25% · 0/20
hip circumference	0.05% · 19/20	4.73% · 0/20	49.79% · 0/20
thigh circumference (L)	0.00% · 20/20	13.32% · 0/20	62.67% · 0/20
calf circumference (L)	0.11% · 19/20	8.08% · 0/20	16.73% · 0/20
arm length (L)	0.00% · 20/20	11.98% · 0/20	16.38% · 0/20
total surface area	0.00% · 20/20	0.00% · 20/20	10.50% · 0/20
total volume	0.00% · 20/20	0.00% · 18/20	0.73% · 1/20

Three rows behave differently. **Height** and **total surface area** are identical for segmentation as well as the port, and **volume** nearly so — they are functionals of the mesh alone, with no landmark in them. Every row that requires deciding *where* diverges. That boundary is the result: the pipelines agree exactly on whatever they do not have to interpret.

What each costs, and the trade space

MATLAB AVATAR SEGMENTATION SLICE

METHOD	MEAN S	MEDIAN S	SLOWEST S	FOLDER S	SCANS/MIN	VS FASTEST
avatar	0.47	0.30	1.31	9	129	1.0×
slice	0.74	0.60	2.03	15	81	1.6×
matlab	3.36	2.55	17.75	67	18	7.2×
segmentation	3.58	3.12	9.10	72	17	7.7×



Mean wall-clock seconds per scan, same folder and same machine. MATLAB excludes Engine start-up, a one-time ~20 s per batch. Slice is measurement only: with plotly installed it also renders seven diagnostic images per subject, which takes it from 0.74 s to 3.0 s.

Three axes, and no method wins on all of them.

- **avatar — 0.47 s/scan, 129 scans/min.** Reproduces MATLAB to floating-point noise at 7.2× less cost than MATLAB itself, with no licence and no Engine install. It inherits every approximation in the reference, including the band-and-hull girth, and emits the fewest columns.
- **slice — 0.74 s/scan.** Essentially free, and the only one computing true cross-sections. The height profile is sound; the reporting layer above it is unfinished.
- **segmentation — 3.6 s/scan, 8× avatar.** Buys the most columns and the only genuinely region-aware girths, and is the only method that has to solve limb separation properly rather than route around it. The folder costs 72 s against avatar's 9 s — which matters at a few hundred scans, not at twenty.

Plainly: if the requirement is *reproduce the historical numbers*, avatar is strictly better than running MATLAB. If the requirement is *measure the body correctly*, none of these is finished, and segmentation is the one whose assumptions sit closest to the anatomy — at roughly twelve times the cost.

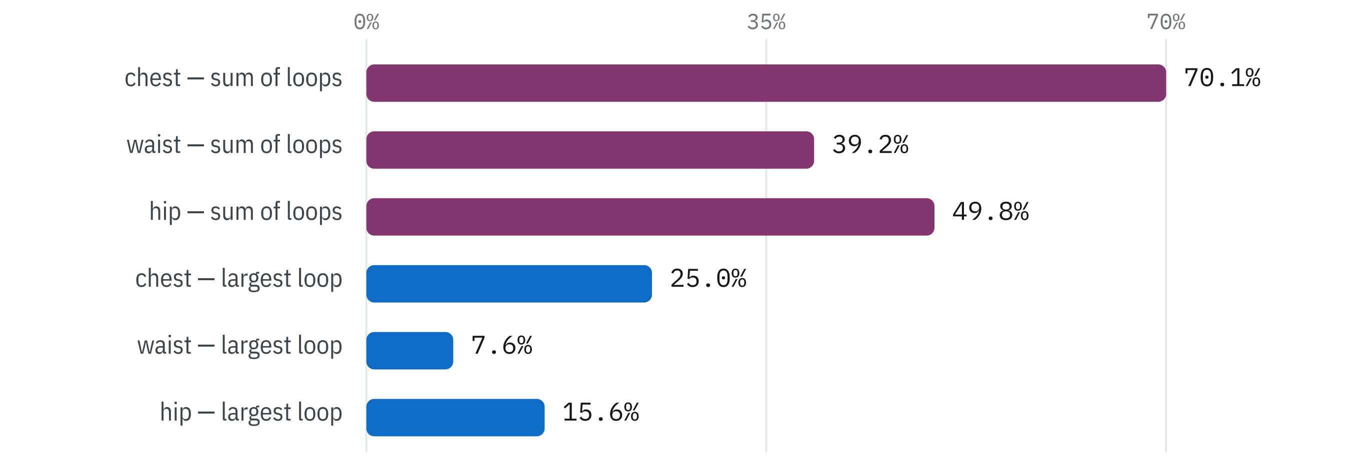
COVERAGE

Measurement columns each backend produces, of 52 canonical columns: **matlab** 42 · **avatar** 36 · **segmentation** 48 · **slice** 42. Breadth and agreement are independent. The port sits six columns below MATLAB because the segment volumes come from a partial hole-filling routine that is not ported; whole-body volume and every segment area are.

Slice: what is unfinished SLICE

> **The geometry is sound; the reporting layer is not.** The backend computes a full height profile per subject — every level, loop count, per-loop perimeter and area, left/right splits — and writes it out. The selections made on top of that are the gap.

- **Aggregation.** A reported girth is `sum_perimeter`. `max_perimeter` sits in the same file. Switching moves chest from 70% to 25% and waist from 39% to 8%, with no change to the geometry underneath.
- **Loop separation.** On 7 of 20 scans the arms touch the torso, so $K = 1$ and the largest-loop rule cannot help either. Those need the loops cut apart.
- **Landmarks.** Stature runs ~2.9 cm short and waist/hip levels are sometimes inverted — both upstream of the aggregation question.



The same levels, aggregated two ways. Upper three bars: what the pipeline reports today. Lower three: what its own profile already contains. The residual is concentrated in the scans where loops merge.

Avatar: the port now reproduces MATLAB MATLAB AVATAR

The avatar backend is a port of `Avatar.m`, so it should return MATLAB's numbers exactly. It now matches 97.4% of 720 paired values, up from 65.4%, after four fixes. None was arithmetic:

CAUSE	WHERE	EFFECT
Do-while trim loops	adjustCrotch	MATLAB drops an element *before* testing, at both ends. Testing first left one extra element and shifted the whole notch profile.
k-means modelling	adjustCrotch	The port solved 2-means exactly. MATLAB seeds randomly and stops at a *local* optimum — frequently not the global one.
Hull loop start	getCircumference → getWrist	`boundary` starts at the lowest-numbered point and repeats it to close the loop. `getWrist` averages that loop, so the duplicated point moved the wrist centroid anc
Face winding	fixFaceOrientation2	Never ported. Two scans carry the same triangle twice with opposite winding; unreconciled, the two signed volumes cancel.

19 values still differ, 18 of them on the single scan above, where the reference's own arm search fails and MATLAB's randomly seeded `kmeans` is not reproducible. The last is a calf girth off by five micrometres.

Reproducing this

```
python -m unified obj2anthro --input data/obj --method avatar --units auto --out runs/avatar
python -m unified.compare runs/methods_report/combined_measurements.csv --reference matlab
python -m pytest unified/obj2anthro/tests/test_avatar_matches_matlab.py
python -m unified.obj2anthro.geometry_figures data/obj --out runs/methods_report/figures --scale-to-cm 0.1
python -m unified.obj2anthro.build_geometry_report runs/methods_report
```

MATLAB values are the recorded run in `runs/matlab_ground_truth/`: `Avatar.m`, `steps=3`, `Vol_SA=on`, MATLAB 23.2.0.2515942 (R2023b) Update 7, via the MATLAB Engine for Python on CPython 3.10. One mesh of 21 (`man.obj`) fails inside `Avatar.m` and is excluded from every comparison. Python backends were re-run over the same folder for this report.

Tables beside this page in `runs/methods_report/`: `combined_measurements.csv`, `comparison_detail.csv`, `comparison_by_measurement.csv`, `comparison_by_subject.csv`, `comparison_coverage.csv`, `girth_decomposition.json`, `report_data.json`.